EASTERN UNIVERSITY
Department of Computer Science and Engineering (CSE)

DATABASE MANAGEMENT LABORATORY

ORCUS

Organized Crime Understanding System

A database-based investigation and evidence management platform

Course	06123228 — Database Management Laboratory
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Team Members

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Proposal Summary

ORCUS is a moderate-scope university DBMS project that organizes agency branches, officers, access roles, GDs, FIRs, cases, separate participant records, locations, evidence, and evidence status history. The design keeps the original academic investigation flow while removing unnecessary operational complexity.

14 conceptual entities • Chen ER notation • 3NF target • Six-week delivery

Background, Problem & Objectives

Abstract

Organized-crime investigations create connected records about complaints, general diaries, FIRs, cases, officers, suspects, victims, witnesses, locations, and evidence. When these records are maintained separately, information becomes difficult to verify and search. ORCUS proposes a centralized relational database for the fictional ORCUS Investigation Agency. The system follows a flexible GD–FIR–Case workflow, maintains separate participant entities, records evidence and its status history, and controls access through users and roles. MySQL will provide the database, Go will implement APIs and business rules, and Next.js will provide the interface. The project is intentionally limited to a moderate academic scope while demonstrating entities, attributes, keys, cardinalities, optional participation, weak entities, M:N relationships, normalization, integrity constraints, history tracking, indexing, and transactional updates.

1. Background

The ORCUS Investigation Agency is a fictional organization that receives complaints, records GDs and FIRs, opens cases, assigns officers, and collects evidence. A structured DBMS is required so that every record has a clear identity and valid relationship to other records.

2. Problem Statement

- Case information may be duplicated across separate files.
- Links among GDs, FIRs and cases may be incomplete or inconsistent.
- Suspect, victim, witness and complainant data are difficult to trace across cases.
- Evidence status changes are not preserved as a reliable history.
- Manual searching and reporting take unnecessary time.

3. General Objective

To design and develop a normalized relational system that manages organized-crime investigation records and demonstrates standard DBMS concepts through a realistic academic prototype.

4. Specific Objectives

- Model 12–14 conceptual entities using Chen notation.
- Support optional GD → FIR and FIR → Case relationships.
- Maintain separate complainant, suspect, victim and witness records.
- Track evidence from registration through status changes.
- Apply users and roles for controlled access.
- Implement CRUD, search, validation and basic reports.
- Convert the conceptual model into a 3NF MySQL schema.

Scope, Modules & DBMS Concepts

1. In-Scope Modules

- Agency branches, officers and officer assignment
- USER, ROLE and M:N user-role access mapping
- Complainant, GD, FIR and Case intake records
- Separate Suspect, Victim and Witness entities
- Case relationships with people and locations
- Weak EVIDENCE entity under CASE
- EVIDENCE_STATUS_HISTORY for auditable changes
- CRUD, search, filters and case summary reports

2. Explicitly Out of Scope

- Native mobile application
- Court workflow or national database integration
- Artificial intelligence, facial recognition or biometrics

3. Functional Requirements

- Create, update, view and search all core records.
- Assign officers and authorized user roles.
- Link a GD to zero or more FIRs and a FIR to zero or more cases.
- Connect cases to multiple suspects, victims, witnesses and locations.
- Prevent orphan evidence and preserve every status change.

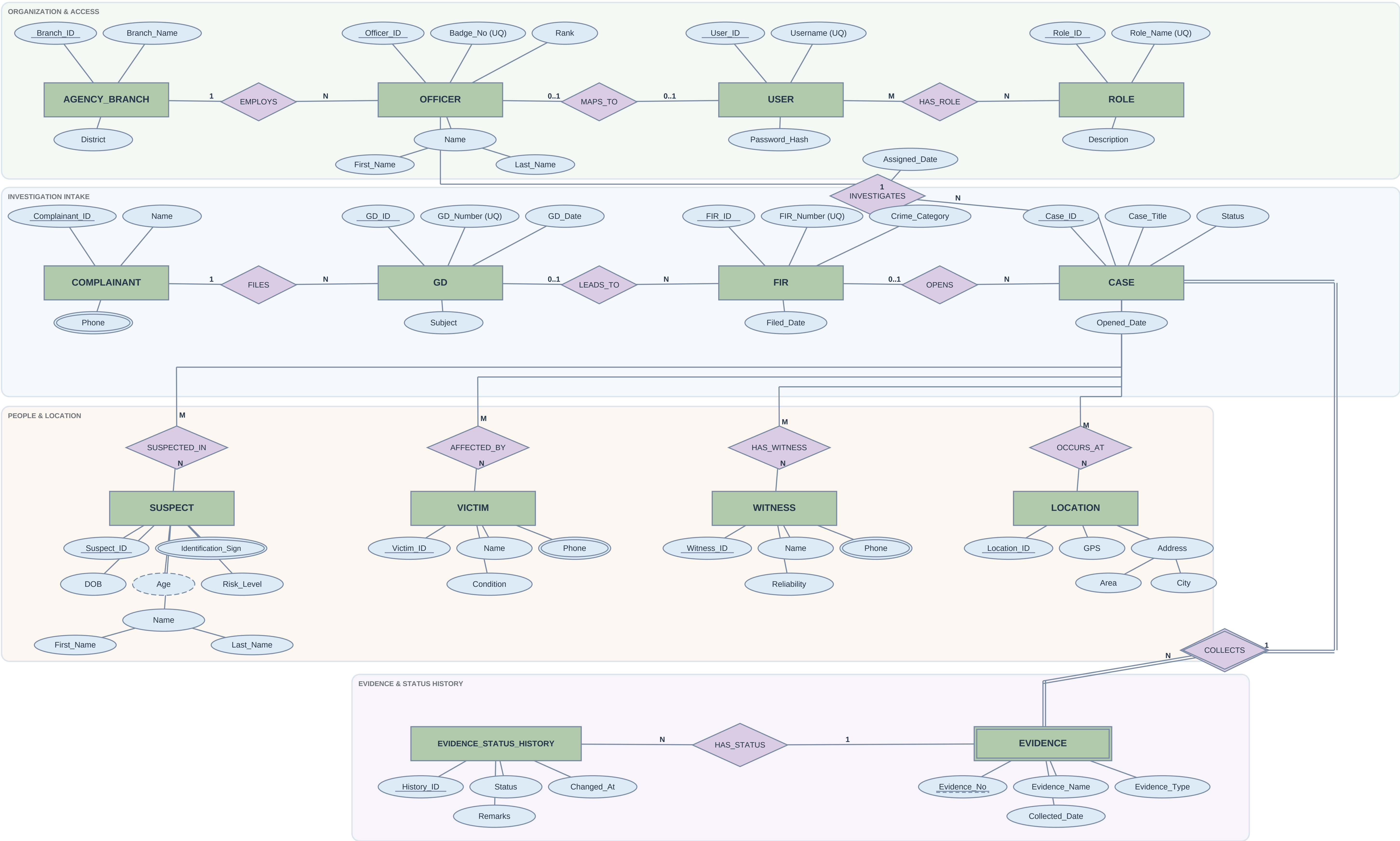
4. Conceptual Entity Inventory

Domain	Entities
Organization	AGENCY_BRANCH, OFFICER, USER, ROLE
Intake	COMPLAINANT, GD, FIR, CASE
People & place	SUSPECT, VICTIM, WITNESS, LOCATION
Evidence	EVIDENCE, EVIDENCE_STATUS_HISTORY

5. DBMS Concepts Demonstrated

Strong entities	Branch, Officer, Case, participant entities
Weak entity	EVIDENCE identified by Case_ID + Evidence_No
Attributes	Key, composite, multivalued and derived
Cardinality	1:1, 1:N and M:N relationships
Participation	Mandatory and optional relationships
Bridge tables	USER_ROLE and case-participant mappings
Normalization	Functional dependencies; target 3NF
Integrity	PK, FK, UNIQUE, CHECK and NOT NULL
History	Append-only evidence status records
Performance	Indexes for case number, status and names
Transactions	Atomic case/evidence updates with rollback

Conceptual Entity–Relationship Diagram



Legend: rectangle = strong entity • double rectangle = weak entity • ellipse = simple attribute • double ellipse = multivalued • dashed ellipse = derived

Underlined = key • dashed underline = partial key • composite attribute has component ellipses • diamond = relationship • double diamond/line = identifying relationship & total participation

Technology, Timeline & Deliverables

1. Technology Stack

Layer	Technology
Database	MySQL / MariaDB
Backend	Go (REST APIs)
Frontend	Next.js
Tools	XAMPP, Git, Draw.io

2. Six-Week Timeline

Week 1 Requirements & final ER

Week 2 MySQL schema & constraints

Week 3 Go APIs: access and intake

Week 4 People, location and evidence

Week 5 Next.js UI, search and reports

Week 6 Testing, documentation and demo

3. Key Deliverables

- Approved five-page project proposal
- Editable Chen ER diagram in Draw.io format
- Normalized MySQL schema and sample dataset
- Go REST API with validation and transactions
- Next.js interface with CRUD, search and reports
- Testing record, source code and final demonstration

4. Business & Integrity Rules

- Every FIR references at most one source GD; the link is optional.
- Every case references at most one source FIR; the link is optional.
- Evidence cannot exist without a case and uses a case-level sequence number.
- Evidence status history is append-only and time-stamped.
- User permissions are assigned through one or more roles.
- Deletion is restricted when dependent investigation records exist.

Expected Outcome & Conclusion

The project will deliver a working academic prototype that demonstrates a complete path from conceptual design to normalized implementation. ORCUS remains achievable within six weeks because it focuses on investigation intake, participant records, locations, evidence, status history, access control, search, and reporting. The design is broad enough to demonstrate important DBMS concepts without adding mobile, court, national-integration, AI, or biometric complexity.

Success measure: correct relationships, valid constraints, reliable history, searchable records, and a clear final demonstration.